

Statistical Report

Maximum Likelihood Estimation and Bootstrap Confidence Intervals for the Generalized Exponential Distribution

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Distribution and Log-Likelihood Function

The dataset was generated from the **Generalized Exponential (GE)** distribution with probability density function

$$f(x; \alpha, \lambda) = \alpha \lambda e^{-\lambda x} (1 - e^{-\lambda x})^{\alpha-1}, \quad x > 0, \alpha > 0, \lambda > 0. \quad (1)$$

For a random sample $x_1, x_2, \dots, x_n$ the log-likelihood is

$$\ell(\alpha, \lambda) = n \ln \alpha + n \ln \lambda - \lambda \sum_{i=1}^n x_i + (\alpha - 1) \sum_{i=1}^n \ln(1 - e^{-\lambda x_i}). \quad (2)$$

The score equations $\partial \ell / \partial \alpha = 0$ and $\partial \ell / \partial \lambda = 0$ do not admit closed-form solutions, so we maximize (2) numerically via the Newton–Raphson algorithm.

Newton–Raphson Algorithm

Gradient (Score Vector)

$$\frac{\partial \ell}{\partial \alpha} = \frac{n}{\alpha} + \sum_{i=1}^n \ln(1 - e^{-\lambda x_i}), \quad \frac{\partial \ell}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i + (\alpha - 1) \sum_{i=1}^n \frac{x_i e^{-\lambda x_i}}{1 - e^{-\lambda x_i}}.$$

Hessian Matrix

Define $A_i = \ln(1 - e^{-\lambda x_i})$, $B_i = x_i e^{-\lambda x_i} / (1 - e^{-\lambda x_i})$, and $D_i = x_i^2 e^{-\lambda x_i} / (1 - e^{-\lambda x_i})^2$. Then

$$H = \begin{pmatrix} -\frac{n}{\alpha^2} & \sum_{i=1}^n B_i \\ \sum_{i=1}^n B_i & -\frac{n}{\lambda^2} - (\alpha - 1) \sum_{i=1}^n D_i \end{pmatrix}.$$

Update Rule and Numerical Safeguards

The update at iteration k is $\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} - s^{(k)} H^{-1} \nabla \ell$, where the step size $s^{(k)}$ is halved (up to 50 times) until $\ell(\boldsymbol{\theta}^{(k+1)}) > \ell(\boldsymbol{\theta}^{(k)})$. A ridge $10^{-6} I$ is subtracted from H for stability; if the solve fails, the algorithm falls back to a gradient-ascent step. Positivity of (α, λ) is enforced at every candidate point.

Initial Guess Values

Six sets of starting values were tried (with $\bar{x}$ denoting the sample mean):

Set	α_0	λ_0
1	1	$1/\bar{x}$
2	2	$1/\bar{x}$
3	3	$1/\bar{x}$
4	2	0.5
5	3	1.0
6	4	1.5

Rationale for initial value selection. For sets 1–3, $\lambda_0 = 1/\bar{x}$ is the MLE of the rate parameter under the simpler Exponential model (i.e. $\alpha = 1$), providing a data-driven and theoretically motivated starting point for λ . The corresponding $\alpha_0 \in \{1, 2, 3\}$ explores a range from the exponential special case upward, since a shape above 1 is plausible for positively skewed lifetime data. Sets 4–6 use fixed round values of λ_0 spanning the plausible range $[0.5, 1.5]$, with $\alpha_0 \in \{2, 3, 4\}$ to probe whether heavier-tailed shapes alter the convergence path. Choosing $\alpha_0 > 0$ and $\lambda_0 > 0$ throughout respects the parameter space and avoids degenerate starting points.

Stopping Criterion

Convergence is declared when *both* of the following conditions hold:

1. **Parameter change:** $\max(|\alpha^{(k+1)} - \alpha^{(k)}|, |\lambda^{(k+1)} - \lambda^{(k)}|) < \varepsilon_1 = 10^{-8}$.
2. **Gradient norm:** $\max(|\partial\ell/\partial\alpha|, |\partial\ell/\partial\lambda|) < \varepsilon_2 = 10^{-4}$.

A maximum of 500 iterations is allowed as a safety cap.

MLE Results and Convergence

All six starting points converged to the same MLE, confirming the uniqueness of the maximum. The results are summarised below.

Init Set	$\hat{\alpha}$	$\hat{\lambda}$	Log-lik	Iterations
1 ((1, $1/\bar{x}$))	3.8798	1.6035	−24.7141	9
2 ((2, $1/\bar{x}$))	3.8798	1.6035	−24.7141	6
3 ((3, $1/\bar{x}$))	3.8798	1.6035	−24.7141	8
4 ((2, 0.5))	3.8798	1.6035	−24.7141	8
5 ((3, 1.0))	3.8798	1.6035	−24.7141	8
6 ((4, 1.5))	3.8798	1.6035	−24.7141	5

Maximum Likelihood Estimates:

$$\hat{\alpha} = 3.8798, \quad \hat{\lambda} = 1.6035, \quad \ell(\hat{\alpha}, \hat{\lambda}) = -24.7141.$$

The algorithm converged in as few as **5 iterations** (starting set 6) and at most **9 iterations** (starting set 1), demonstrating rapid convergence regardless of the initial values.

Bootstrap Confidence Intervals

Both methods used $B = 500$ bootstrap replications.

4.1 Parametric Bootstrap

New samples of size $n = 25$ were drawn from the fitted $\text{GE}(\hat{\alpha}, \hat{\lambda})$ distribution using the inverse-CDF method:

$$X = -\frac{1}{\hat{\lambda}} \ln(1 - U^{1/\hat{\alpha}}), \quad U \sim \text{Uniform}(0, 1).$$

For each replicate the NR algorithm was re-run from the MLE as starting point, giving bootstrap estimates $(\hat{\alpha}_b^*, \hat{\lambda}_b^*)$, $b = 1, \dots, 500$. The 95% confidence intervals are the 2.5th and 97.5th empirical percentiles:

Parameter	MLE	95% CI Lower	95% CI Upper
α	3.8798	2.1927	12.8089
λ	1.6035	1.1346	2.5488

4.2 Non-Parametric Bootstrap

Bootstrap samples were drawn by re-sampling $n = 25$ observations *with replacement* from the original dataset. The NR algorithm was applied to each resample in the same way.

Parameter	MLE	95% CI Lower	95% CI Upper
α	3.8798	2.7206	8.2768
λ	1.6035	1.1956	2.5496

4.3 Comparison and Discussion

The non-parametric CIs for α are narrower than the parametric ones (width ≈ 5.56 vs. ≈ 10.62): the parametric method propagates additional uncertainty from the assumed model's tails, while the non-parametric method is anchored to the observed data. The CIs for λ agree closely across both methods (widths 1.414 and 1.354), confirming that λ is estimated with greater precision than α at $n = 25$.

The upper-skewed intervals for α reflect the well-known positive skewness of shape-parameter estimates in lifetime models at moderate sample sizes. The parametric bootstrap assumes GE model correctness; the close agreement of the λ intervals across both methods lends informal support to this assumption.

Summary

- **File number received:** `fortdataset.36` ($n = 25$ observations).
- **MLEs obtained:** $\hat{\alpha} = 3.8798$, $\hat{\lambda} = 1.6035$ with $\ell(\hat{\alpha}, \hat{\lambda}) = -24.7141$.
- **Initial guesses:** Six sets of (α_0, λ_0) were tried; all converged to the same MLE.

- **Stopping criterion:** Maximum absolute parameter change $< 10^{-8}$ AND maximum absolute score component $< 10^{-4}$.
- **Iterations:** Convergence achieved in 5–9 iterations depending on starting point.
- **Bootstrap replications:** $B = 500$ for both parametric and non-parametric methods.